

# What Roles do Drugs Play in Coercion and Control?



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Drugs are also used to control individuals. The story of the calabar bean presented by Balick and Cox is a vivid example of how elite members of society maintain control through dangerous drugs. We find a different social circumstance in which drugs were used in a ritual setting, which also involved elite control of individuals, in the Inca culture of South America. Natural mummies, preserved by extremely cold and dry conditions at very high elevations in the Andes mountains, include hair, fingernails, and other tissues uncommonly preserved archaeologically. The hair of these individuals documents their use of narcotics, including the presence of both coca and alcohol in a sacrificed young woman of the Inca period.

The Inca believed that it was sometimes necessary to leave individuals, often children or young women, to die of exposure (occasionally, they were first killed more violently) on top of mountains as an offering to the spirits of ancestors and the mountains themselves—a ritual termed *capachocha*. These bodies became naturally mummified by the extremely cold and dry conditions in these areas. The unique circumstances of preserving these mummies, with long, intact hair, made it possible to determine how long before death the use of both coca and alcohol began. A study by [Andrew Wilson and colleagues \(https://www.pnas.org/doi/full/10.1073/pnas.1305117110\)](https://www.pnas.org/doi/full/10.1073/pnas.1305117110) documents the life history of a particularly well preserved, sacrificed young woman (roughly 13 years old at death), using a combination of CT scanning—which enables the visualization of her skeleton and other internal organs without invasive techniques—and chemical analyses of both residues and stable isotopes in her hair.

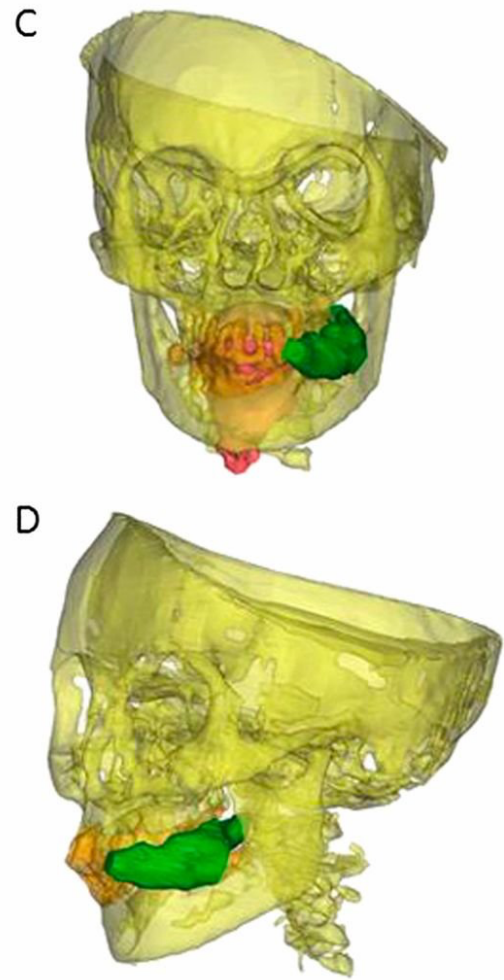
The following contains images of mummified human remains from an archaeological excavation. It is not necessary to view these images to complete this course.

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☒ A. **Correct:** Yes, I would like to view these images.

Correct answer

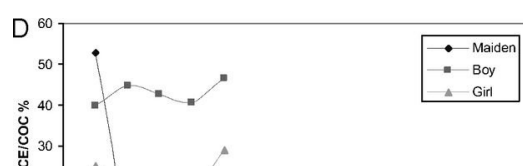
☐ B. No, I would **not** like to view these images.

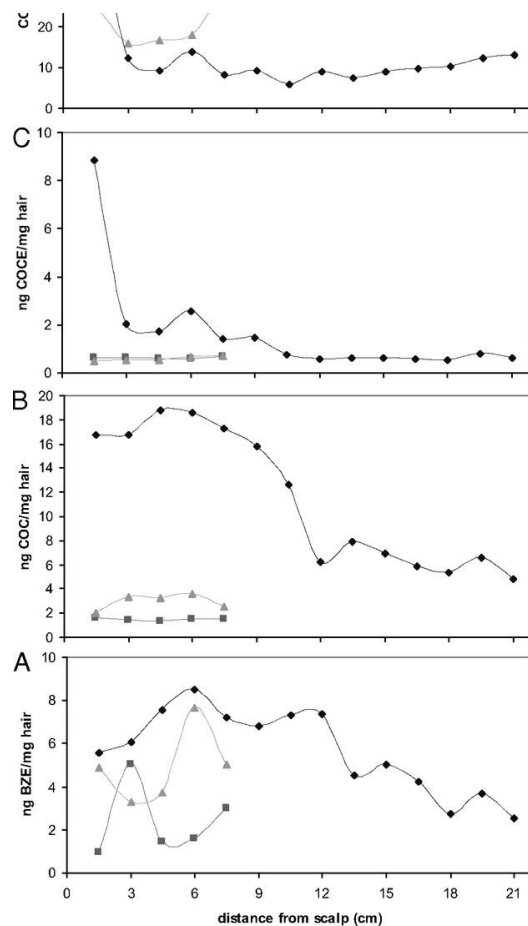


Face of the Inca woman sacrificed and naturally mummified, when partially unwrapped (A). CT visualization of a quid (wad of chewed) coca in her left cheek, shown in green (B-D); its slight bulge can also be seen in A.

Source: Fig. 4 in Wilson, A. S., et al. (2013). Archaeological, radiological, and biological evidence offer insight into Inca child sacrifice. *Proceedings of the National Academy of Sciences*, 110(33), 13322-13327. <https://doi.org/doi:10.1073/pnas.1305117110>

A wad of chewed coca is still present in her mouth; it was being chewed at the time of death. How long had this young woman been chewing coca? That was determined by the study's authors using her hair. Human hair (and fingernails) record traces of metabolized chemicals that are incorporated in trace amounts. Since our hair and nails grow over time, the far ends of these tissues record the chemical levels present when those cells formed. When an individual with long hair, such as the young woman here, is studied, a length of hair may record one or more years of diet and chemical residue, if the hair is analyzed in small pieces from the scalp (recent) to the end of the hair (oldest). The authors did so and present their results in this diagram; her hair represents approximately 2 years of growth.





Four different chemicals, all representing metabolites of coca, present in the hair of the sacrificed young woman (black diamonds), compared with two other younger (4–5-year-old) sacrificed children. The x-axis in all graphs is distance from the scalp, which represents time before death, with most recent to the left.

Source: Fig. 5 in Wilson, A. S., et al. (2013). Archaeological, radiological, and biological evidence offer insight into Inca child sacrifice. *Proceedings of the National Academy of Sciences*, 110(33), 13322-13327.  
<https://doi.org/doi:10.1073/pnas.1305117110>

The authors looked at four distinct measures of coca use. The lower two on this graph (A, B) represent two primary metabolites of coca: COC = cocaine, BZE = benzoecgonine, a metabolite of cocaine. Graph C maps the metabolite cocaethylene (COCE), a metabolite of cocaine formed only in the co-presence of alcohol. Graph D compares COCE to COC to look at the presence of alcohol independent of cocaine use.

The interpretation of these metrics helps to inform the last two years of the young woman's life. There was a significant increase in her consumption of coca beginning about one year before her death (at 9-12 cm from scalp, both BZE and COC increased). In contrast, alcohol use (as measured

by COCE, graphs C and D) began at low levels around the same time but increased significantly shortly before death, in the final weeks of life.

The authors conclude: "It is likely that coca and alcohol played a dual role at the end of these sacrificial victims' lives. Understood within the cultural frameworks of Inca religious ideology, both were associated with elite ritual practice. Coca and alcohol were substances that induced altered states interpreted as sacred, and which could suggest to victims and those associated with them the proximity of the divine beings whose continued benevolence was underwritten by these rites. From a cross-cultural perspective, the psychologically deadening, disorienting, and mood-modifying effects of these psychoactive compounds on young victims, for whom any kind of informed consent to their own deaths cannot be unproblematically presumed, should not be downplayed" (Wilson et al. 2013, p. 13326). In short: the combined use of coca and alcohol was likely a deliberate strategy used by Inca elite in controlling young children during the rituals that